LECTURE 1

1. Simple extensions

Let K be a field and let $a \in K$. Recall that the *minimal polynomial* of a over K is the monic polynomial $m(x)$ of smallest degree such that $m(a) = 0$.

Lemma 1.1. (On minimal polynomials) *Suppose that K is a subfield of L, and $a \in L$ is algebraic over K.*

(a) Let m be a minimal polynomial of a over K, i.e. $m \in K[t]$ is a monic polynomial of minimal possible degree such that $m(a) = 0$. Then m is irreducible over K. For $f \in K[t]$ we have $f(a) = 0$ if and only if $m \mid f$. It follows that the minimal polynomial is unique.

(b) Conversely, if $m \in K[t]$ is a monic irreducible polynomial such that $m(a) = 0$, then m is the minimal polynomial of a.

Proof. (a) If m was reducible, then $m = m_1 m_2$ for some monic m_1, m_2 of smaller degree. Substituting a we get $0 = m(a) = m_1(a)m_2(a)$, which implies that either $m_1(a) = 0$ or $m_2(a) = 0$, since fields have no zero divisors. This is, however, a contradiction, since m was of smallest degree.

Let $f(a) = 0$. Write $f = mq + r$ with $\deg r < \deg m$; we need to show that $r = 0$. We have

$$0 = f(a) = m(a)q(a) + r(a) = 0 \cdot q(a) + r(a) = r(a).$$

If $r \neq 0$, then from the equation $r(a) = 0$ and multiplying by a suitable constant from K if necessary, we obtain a monic polynomial in $K[t]$ having a as a root of degree smaller than $\deg m$, a contradiction. Conversely, if $m \mid f$, then $f = mq$, whence $f(a) = m(a)q(a) = 0 \cdot q(a) = 0$.

It follows that the minimal polynomial is unique, for if m' is another one, we have both $m \mid m'$ and $m' \mid m$ (as proved above), which implies $m = m'$.

(b) Let u be the minimal polynomial of a. Then $u \mid m$ by (a), as $m(a) = 0$. However if m is irreducible, it follows that $u = m$ since both are monic. □

Theorem 1.2. *If m is an irreducible polynomial of degree greater than 0 over a field K, then the factor-ring $L = K[t]/I$ is a field, where I is the ideal of $K[t]$ generated by m (the set I actually consists of all polynomials that are divisible by m). The image of K, i.e. the set of the cosets $k + I$, for $k \in K$, is a subfield of L isomorphic to K. The image of t, i.e. the coset $t + I$, is a root of m.*

Proof. Since $K[t]$ is an associative commutative ring with 1, so is the quotient ring $K[t]/I$. In particular, the coset $1 + I$ is the unity in $K[t]/I$. We need only show that every non-zero element $a + I$ has a multiplicative inverse. To be non-zero is equivalent to $a \notin I$, i.e. $m \nmid a$. Since m is irreducible this implies that $(m, a) = 1$. We know that there are $u, v \in K[t]$ such that $um + va = (m, a) = 1$. Taking the images of this equation under the natural homomorphism of $K[t]$ onto $K[t]/I$ gives that the image of v is the inverse of that of a, since the image of $um \in I$ is zero. In other words, we have $(v + I)(a + I) = va + I$ by definition, which equals $1 - um + I$ by the above. However $um \in I$, so $1 - um + I = 1 + I$, the unity of $K[t]/I$. Thus $v + I$ is the required inverse of $a + I$:

$$(v + I)(a + I) = va + I = 1 - um + I = 1 + I.$$

Hence we have proved that L is a field.

The mapping $k \mapsto k + I$ of K onto L is a injection:

$$k + I = k' + I \quad \Longrightarrow \quad k - k' \in I \quad \Longrightarrow \quad m \mid (k - k') \quad \Longrightarrow \quad k - k' = 0,$$

since $\deg m > 0$. This mapping is a homomorphism - this easily follows from the definition of the operations on $K[t]/I$. Thus, the image of K is a subfield of L isomorphic to K.

Now let $a = t + I$ be the image of t in $K[t]/I$. The definition of the operations in the quotient ring imply that $m(a) = m(t + I) = m(t) + I$. The right-hand side is zero, since $m(t) \in I$. □

Corollary 1.3. (Existence of simple extensions) *For every (monic) irreducible polynomial $m \in K[t]$ over a field K there is a simple algebraic extension $K(a)$ such that m is the minimal polynomial of a over K.*

Proof. Put $a = t + I$ in $L = K[t]/I$ as in the theorem above. Then $L = K(a)$, since every element of L has the form $f(t) + I = f(t + I) = f(a)$ for $f \in K[t]$. As we saw, $m(a) = 0$. Since m is irreducible, we have that m is the minimal polynomial of a over K by the lemma above. □

Recall that an extension $L : K$ is *algebraic* if every element of L is algebraic over K.

Theorem 1.4. (Description of simple algebraic extensions) *Let $K(a)$ be a simple algebraic extension of a field K and let m be the minimal polynomial of a over K. Then every element of $K(a)$ has a unique representation in the form $f(a)$ for $f \in K[t]$ with $\deg f < \deg m$. (These f need not be monic, of course.)*

Proof. First, we claim that $K(a) = \{g(a)/h(a) \mid g, h \in K[t],\ h(a) \neq 0\}$. Indeed, it is easy to see that the right-hand side is closed under all operations and hence is a (sub)field. By definition, the set $K(a)$ is the smallest subfield containing a, hence $K(a) \subseteq$ RHS, hence equality follows. Now for every such element $h(a) \neq 0$ we have that $m \nmid h$ (see above lemma). Since m is irreducible this implies that $(h, m) = 1$. Then there are $u, v \in K[t]$ such that $uh + vm = 1$. Hence $u(a)h(a) + v(a)m(a) = 1$, whence $(h(a))^{-1} = u(a)$ since $m(a) = 0$. As a result,

$$g(a)(h(a))^{-1} = g(a)u(a) = w(a),$$

where $w = gu \in K[t]$. Write $w = mq + r$ with $\deg r < \deg m$. Then

$$w(a) = m(a)q(a) + r(a) = r(a),$$

since $m(a) = 0$. Thus, every element of $K(a)$ can be represented in the required form.

To show uniqueness, suppose that $f_1(a) = f_2(a)$ with $\deg f_1, \deg f_2 < \deg m$. Then $(f_1 - f_2)(a) = 0$, whence $m \mid (f_1 - f_2)$ by the lemma above. This implies $f_1 - f_2 = 0$, since the degree is smaller than that of m. □

Corollary 1.5. (Degree of simple algebraic extensions). *Let $K(a)$ be a simple algebraic extension of a field K and let m be the minimal polynomial of a over K. Then $[K(a) : K] = \deg m$.*

Proof. Recall that the degree $[L : K]$ is the dimension of L regarded as a vector space over K. The above theorem actually states that 1, a, a^2, $\ldots$, a^{n-1} is a basis of $K(a)$ over K, where $n = \deg m$. Indeed, a linear combination of these elements with coefficients b_0, b_1, $\ldots$, b_{n-1} is precisely the value of the polynomial $b_0 + b_1 t + \cdots + b_{n-1}t^{n-1}$ on a. Now the uniqueness statement obviously amounts to the linear independence. □

Theorem 1.6. (Finite extensions) (a) *If an extension $L : K$ is finite (i.e. $[L : K]$ is finite), then $L : K$ is algebraic and $L = K(a_1, \ldots, a_s)$ (i.e. L can be obtained from K by adjoining finitely many elements).*

(b) *If $L = K(a_1, \ldots, a_s)$ and every a_i is algebraic over K, then $[L : K]$ is finite.*

Proof. (a) Let $[L : K] = n$. Then L has a finite basis $a_1, \ldots, a_n$ as a vector space over K. Then of course $L = K(a_1, \ldots, a_n)$. Let a be any element of L. The $n + 1$ elements 1, a, a^2, $\ldots$, a^n must be linearly dependent in the n-dimensional K-space L. This means that there exist coefficients b_0, b_1, $\ldots$, $b_n \in K$ (not all zero) such that $b_0 + b_1 a + \cdots + b_n a^n = 0$. Now the LHS is precisely the value of the (non-zero!) polynomial $b_0 + b_1 t + \cdots + b_{n-1} t^{n-1}$ on a. Thus a is algebraic over K by definition.

(b) Consider the tower of extensions:

$$K \subseteq K(a_1) \subseteq K(a_1, a_2) \subseteq \cdots \subseteq K(a_1, \ldots, a_{i-1}) \subseteq K(a_1, \ldots, a_i) \subseteq \cdots \subseteq K(a_1, \ldots, a_s) = L.$$

Every step is a simple extension: $K(a_1, \ldots, a_i) = K(a_1, \ldots, a_{i-1})(a_i)$, with a_i being adjoined to the field $K(a_1, \ldots, a_{i-1})$. Since a_i is algebraic over K, it is algebraic over $K(a_1, \ldots, a_{i-1})$. If m_i is the minimal polynomial of a_i over $K(a_1, \ldots, a_{i-1})$, then

$$[K(a_1, \ldots, a_i) : K(a_1, \ldots, a_{i-1})] = \deg m_i.$$

Now, by the Tower Law, the degree $[L : K]$ is the product of s (finite) numbers $\deg m_i$, so $[L : K]$ is finite, as required. □

Definition 1.7. A *field isomorphism* is a ring isomorphism between two fields. Likewise, a *field homomorphism* is a ring homomorphism between two fields.

Example 1.8. (See Exercise Sheet 1 for details.) In $\mathbb{Z}_3[x]$, let I be the principal ideal generated by the irreducible polynomial $x^3 - x + 1$ and let J be the principal ideal generated by the irreducible polynomial $x^3 - x - 1$. Then $\theta : \mathbb{Z}_3[x]/I \to \mathbb{Z}_3[x]/J$ defined by $f(x) + I \mapsto f(x) + J$ is a bijection but is not a field isomorphism. However $\phi : \mathbb{Q}(e) \to \mathbb{Q}(\pi)$ defined by $f(e) \mapsto f(\pi)$ for any $f(x) \in \mathbb{Q}[x]$ is a field isomorphism. Similar to the first map, the map $\theta : \mathbb{Q}(\sqrt{2}) \to \mathbb{Q}(\sqrt{5})$ defined by $a + b\sqrt{2} \mapsto a + b\sqrt{5}$ is not a field isomorphism.

Now suppose $i : K \to K'$ is a field isomorphism. It is then natural to extend i to an isomorphism of polynomial rings $i : K[t] \to K'[t]$, where for $f \in K[t]$ we obtain $i(f) \in K'[t]$ by applying i to all coefficients of f: if $f = b_0 + b_1 t + \cdots + b_{n-1} t^{n-1}$, then $i(f) = i(b_0) + i(b_1)t + \cdots + i(b_{n-1})t^{n-1}$. This extended i is now an isomorphism of $K[t]$ with $K'[t]$.

Example 1.9. Using the isomorphism $\phi : \mathbb{Q}(e) \to \mathbb{Q}(\pi)$, we then have an isomorphism of polynomial rings $\mathbb{Q}(e)[t] \to \mathbb{Q}(\pi)[t]$, as defined above.

Theorem 1.10. (Isomorphisms of simple algebraic extensions) *Suppose*

- *$i : K \to K'$ is a field isomorphism;*
- *$K(a)$ is an extension of K with m being the minimal polynomial of a over K; and*
- *$K'(b)$ is an extension of K' with $i(m)$ being the minimal polynomial of b over K'.*

Then there exists an isomorphism $\mu : K(a) \to K'(b)$ such that $\mu(a) = b$ and $\mu|_K = i$.

Proof. Using the description theorem above, we define $\mu(f(a)) = i(f)(b)$, for every $f \in K[t]$ with $\deg f < \deg m$ (we replace all coefficients by their images under i and the powers of a by those of b). The uniqueness statement of the description theorem guarantees that μ is well defined and, moreover, that μ is a bijection of $K(a)$ onto $K'(b)$. It is clear that $\mu(a) = b$ and $\mu|_K = i$. It remains to show that μ preserves all operations. See Exercise Sheet 1. □

Remark 1.11. One can look at the easier case when $K = K'$ and i is the identity mapping, where the above expressions are more transparent, without i. However even if $K' = K$, the mapping i need not be the identity mapping; we shall need this case later, when i is an automorphism of K.

Example 1.12. Consider $\mathbb{Q}(\sqrt[3]{2})$ which is a simple extension of $\mathbb{Q}$ of degree 3, with $t^3 - 2$ being the minimal polynomial of $\sqrt[3]{2}$ (since it is irreducible by Eisenstein's). Let $\omega = e^{2\pi i/3} = -1/2 + i\sqrt{3}/2$ be a complex cubic root of 1. Then $b = \omega\sqrt[3]{2}$ is also a root of $t^3 - 2$. Although $\mathbb{Q}(b)$ is no longer a subfield of reals, the fields $\mathbb{Q}(\sqrt[3]{2})$ and $\mathbb{Q}(b)$ are isomorphic: everything stated about the inner structure of $\mathbb{Q}(\sqrt[3]{2})$ can be "translated" into a statement about the inner structure of $\mathbb{Q}(b)$ by "replacing" $\sqrt[3]{2}$ by b.